Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

HABITAT SURVEY REPORT

February 2017

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1.0 INTRODUCTION

1.1 The following report has been produced by FPCR Environment & Design Ltd on behalf of a consortium comprising Hallam Land Management Ltd, Taylor Wimpey and Hannick Homes to provide details of results of habitat surveys undertaken at land to the East of Swindon Wiltshire.

Site Location and Context

1.2 The Study Area is centred on the Ordnance Survey Grid Reference SU 2007 8752 and lies to the east of Swindon close to the village of South Marston.

1.3 The site is primarily used for agriculture at present, with pastoral fields bisected by hedgerow field boundaries in the west, to the south of South Marston, and a range of agricultural uses centred around Rowborough Farm to the east, including arable use, turf production and biomass fuel production.

1.4 The site and wider area to the south has been the focus of a wide range of surveys from at least 1999, including a number completed by FPCR.

Objectives of Study

1.5 The objectives of the study were to:

- assess the relative interest of the survey area
- where necessary complete further detailed assessment to determine relative nature conservation value
- provide recommendations for mitigation
- review the broad site proposals and opportunities for enhancement

2.0 METHODOLOGY

Desk Study

2.1 In order to compile existing baseline information, relevant botanical information was requested from both statutory and non-statutory nature conservation organisations for the purposes of this appraisal, including:

- Multi Agency Geographic Information for the Countryside (MAGIC) website[1]
- The National Biodiversity Network (NBN) website[2]
- Botanical Society of the British Isles (BSBI) County recorder
- Wiltshire and Swindon Biological Records Centre (WSBRC)

[1] http://www.magic.defra.gov.uk/
[2] http://www.nbn.org.uk/

Flora

2.2 Survey methods followed the extended Phase 1 Survey technique as recommended by Natural England (JNCC, 1990 revised 2003). This involved a systematic walk over of the site by an experienced ecologist initially on 17 June 2013 to classify the habitat types present and marking them on a base map. This survey was then updated in Spring/Summer 2015 and again in January 2017 to ensure that habitat descriptions remained consistent with those described.

2.3 An assessment of the botanical interest of the site was therefore undertaken following the identification of the relative cover of species recorded. This was collected using the DAFOR scale:

D Dominant

A Abundant

F Frequent

O Occasional

R Rare

2.4 The DAFOR scale, although does not have any agreed quantitative values is a useful measure of assigning relative cover to species within as sward, as used, for example by Natural England in assessing Sites of Special Scientific Interest (SSSIs).

3.0 RESULTS

Desk Study

3.1 A range of notable botanical species were highlighted by the BSBI and WSBRC as being present in the wider area. Only three species occur in close proximity to the site: black poplar *Populus nigra subsp. Betulifolia,* Plane toadflax *Linaria repens* and dwarf spurge *Euphorbia exigua*. Only dwarf spurge was found within the site, although this is an old record (1984) for a short lived ephemeral species. A summary is provided in Table 1 below.

Table 1: Notable botanical records in the wider area

Taxon	Category	Site	Grid ref	Comment
Populus nigra subsp. betulifolia	VC7 Rare	Swindon, east of	SU195829	4 trees just north of Great Moor Lease Farm. Low pollards, in excellent health with from 4 - 15 poles per tree. Probably male. Trees are strung out along a ditch that eventually drains into the R. Cole
Populus nigra subsp. betulifolia	VC7 Rare	Swindon	SU196829	N of Little Moor bet. A419 & C road
Populus nigra subsp. betulifolia	VC7 Rare	Swindon, east of	SU195850	1 tree near Wanborough House (west of A419). Low pollard, in excellent health with from 4 - 15 poles per tree. Probably male. Trees are strung out along a ditch that eventually drains into the R. Cole

Taxon	Category	Site	Grid ref	Comment
Populus nigra subsp. betulifolia	VC7 Rare	Marsh Farm	SU196851	2 trees, one either side of Wanborough Rd at crossing with A419
Euphorbia exigua	IUCN Near Threatened list	SU18Y	SU1987	
Linaria repens	VC7 Scarce	South side of railway	SU197871	Many plants.
Populus nigra subsp. betulifolia	VC7 Rare	Acorn Bridge (Nr)	SU215881	Along streamside, 1 tree
Taxus baccata	Wilts Rare Plant List	600m north of boundary	SU1988	WSBRC
Aquilegia vulgaris	Not given	300m north of site	SU195883 –	WSBRC
Pilosella agg	Not given	700m north of site	SU192885 –	WSBRC
Lactuca serriola	Not given	1km northwest of site	SU1888 –	WSBRC
Euphorbia exigua	IUCN Near Threatened	within site	SU1987 –	Sarah Pilkington – Nicolas Pearson Associates
Linaria repens	VC Scarce	South of site	SU197871 –	Sarah Pilkington – Nicolas Pearson Associates
Ruscus aculeatus	Wilts Rare Plant List	200m east of boundary	SU2083	WSBRC
Sanguisorba officinalis	Wilts Rare Plant List	-	SU208867	WSBRC
Sanguisorba officinalis	Wilts Rare Plant List	-	SU194852	WSBRC
Sanguisorba officinalis	Wilts Rare Plant List	2km east of site	SU217854	WSBRC
Solidago virgaurea	Wilts Rare Plant List	2km east of site	SU217854	WSBRC

Field Survey – Habitats

Overview

3.2 A range of habitats predominate within the site, which include extensive areas used for turf and other arable production. Further areas are currently used for hay or cattle/sheep pasture. Field boundaries are dominated by hedgerows, which frequently include mature standard trees.

3.3 The South Marston Brook flows from the north of the site to the east where it joins the River Cole. A small number of ponds also occur scattered within the site.

Broad–leaved plantation woodland

3.4 Broad-leaved plantation woodland occurs to the south of Rowborough Farm where evenly aged drawn willow specimens of 10-15yrs growth occur, including crack *Salix fragilis* and osier willows *S. viminalis* with occasional wych elm *Ulmus glabra* (Photograph 1). The dense canopy provided a high level of shade, with consequently little real understory. The ground flora was of low diversity and generally dominated by nettle *Urtica dioica* with other occasional tall ruderal herbs, although in the south-eastern corner (Target Note (TN) 1) supported a number of species more characteristic of waterlogged ground such as reed sweet grass *Glyceria maxima,* lesser pond sedge *Carex acutiflorus*, common comfrey *Symphytum officinale,* meadowsweet *Filipendula ulmaria* and reed canary grass *Phalaris arundinacea*.

Photograph 1: Established willow coppice with coarse/ruderal field layer

Coniferous Plantation Woodland

3.5 A small Christmas tree plantation dominated by Norway Spruce *Picea abies* is located to the northeast of Rowborough Farm. The trees are densely planted and relatively tall to 5-10m in height, allowing little associated vegetation to persist other than small patches of rough meadow grass Poa trivialis.

Semi-improved Grassland

3.6 Semi-improved grassland occurs scattered to the south of the site and in field compartments south of South Marston and persists where intensive agricultural management is reduced. All are species–poor in nature and where ungrazed, are dominated by a range of common coarse grasses.

3.7 Semi-improved grassland has established adjacent to the railway line in the southeast of the site, where it has developed on former arable farmland part planted with coppice (Photographs 2)..

False-oat grass dominates locally although tufted hair grass Deschampsia cespitosa, cock's-foot Dactylis glomerata and meadow foxtail Alopecurus pratensis are also frequent. Although taller ruderal species generally predominate, the grassland also supports a reasonable diversity of herbs such as oxeye daisy *Leucanthemum vulgare*, meadow vethcling *Lathyrus pratensis*, tufted vetch *Vicia sativa* and common mouse ear *Cerastium fontanumn*, which appears to have colonised from the adjacent railway embankment and appears to be reasonably species rich.

3.8 A small area of semi-improved grassland is associated with a track in the south of the site (TN2). The grassland supports a number of species characteristic of less improved grasslands, although is very limited in extent. Additional species observed included ladies bedstraw Galium verum, common bird's-foot trefoil *Lotus corniculatus*, hoary ragwort *Senecio erucifolius*, black medick *Medicago lupulina*, alsike clover *Trifolium hybridum* and hedge bedstraw *Galium albus*, which were all localised along the track margins.

3.9 Where grazed, such as the small paddocks adjacent to Rowborough Farm and Manor Farm, constant species included perennial ryegrass *Lolium perenne*, Yorkshire fog *Holcus lanatus* and crested dog's-tail *Cynosurus cristatus* with a range of grazing tolerant herbs such as creeping buttercup *Ranunculus repens*, white clover *Trifolium repens*, cut-leaved crane's-bill *Geranium dissectum*, common cat's-ear *Hypochoeris radicata* and greater plantain *Plantago major* (Photograph 3).

Photograph 2: Unmanaged semi-improved coarse grassland adjacent to Acorn Bridge

Photograph 3: Heavily grazed horse pasture close to Rowborough Farm

3.10 Some small field compartments to the south of South Marston also support semi-improved grassland sward. The first is located just to the north of Priory Farm where a small number of pear trees exist. Although some species indicative of less intensive management were noted, such as meadow vetchling and common sorrel *Rumex acetosa*, coarse grasses were overwhelmingly dominant with false oat grass and common couch of high frequency. The diversity and frequency of herbs was low.

3.11 Further fields to south of South Marston (TN 3) support a sward, indicative of past heavy grazing but which is now unmanaged. Red fescue Festuca rubra, rough meadow grass *Poa trivialis* meadow buttercup *Ranunculus acris*, and Yorkshire fog *Holcus lanatus* were abundant, with occasional red clover, selfheal *Prunella vulgaris*, tufted vetch and, rarely meadow barley *Hordium secalinum*.

3.12 Further south still, a sward more typical of damper ground persists (TN 4). Meadow foxtail and creeping bent Agrostis stolonifera were abundant, with frequent rough meadowgrass, meadow buttercup, creeding buttercup and perennial ryegrass *Lolium perenne*. Meadow vetchling is also

very locally frequent Damp depressions supported occasional silverweed *Potentilla anserina* jointed rush *Juncus articul*atus, floating sweet grass *Glyceria fluitans*, hairy sedge *Carex hirta* and marsh foxtail *Alopecurus geniculatus*.

Improved grassland

3.13 Improved grassland occurs frequently in the south of the site in field compartments adjacent to the railway corridor. Perennial ryegrass and rough meadowgrass are generally abundant with a range of frequent associates such as meadow foxtail, timothy *Phleum pratense*. In some areas the sward is becoming less intensively managed and supports occasional false-oat grass, meadow barley, tall fescue *Festuca arundinacea* and Yorkshire fog *Holcus lanatus*.

3.14 Two field compartments supporting improved grassland also occur in the east of the site, where a willow biofuel crop was previously grown. Following harvesting these fields have been sown with grassland, presumably to be used as cattle pasture. Both are characteristically of low species diversity.

Tall Ruderal Vegetation

3.15 Tall ruderal vegetation is located frequently along hedgerows bases as is typical where arable and improved grassland predominate. The hedge bottom flora is largely ruderal in its composition with bramble *Rubus fruticosus* agg., common nettle *Urtica dioica* and cleavers *Galium aparine* all frequent and often abundant throughout the hedge network. Cock's-foot, Yorkshire-fog *Holcus lanatus* and false oat-grass were all key grass species.

3.16 Significant areas of tall ruderal vegetation was located in the south of the site. The first was associated with an area of tipped material (TN 5; Photograph 4) where teasel *Dipsacum fullonum*, nettle *Urtica dioica* and ragwort *Senecio jacobaea* occur in addition to a range of coarse grasses.

Photograph 4: Ruderal Vegetation and tipped material

Water Bodies – Running Water

3.17 Two water courses, the River Cole and a tributary of it, the South Marston Brook, are associated with the survey area. The River Cole exists to the east of the application Site. Its banks are typically steep to around 2m height and supported a marginal flora including occasional marsh woundwort *Stachys palustris*, reed canary grass, gypsywort *Lycopus europaea*, water mint *Mentha aquatica* and amphibious bistort *Polygonum amphibium*. In-channel vegetation included occasional stands of branched bur-reed Sparganium erectum, fool's watercress *Apium nodiflorum* and yellow flag iris *Iris pseudacorus*.

3.18 The South Marston Brook flowed through the Site from South Marston in the north through to its confluence with the River Cole in the east. It is heavily shaded throughout much of its length by adjacent ruderal vegetation, trees and scrub. The banks range from steep to vertical and are undercut in places.

3.19 Tall herb vegetation including abundant nettle, greater willowherb, and coarse grasses typically dominated the bankside where more open in character (Photograph 5). A marginal flora was present including common comfrey, greater willowherb and meadow sweet. An associated ditch supported a more diverse wetland flora including watercress *Rorippa nasturtium-aquatica*, brooklime *Veronica beccabunga*, reedmace *Typha latifolia*, greater pond sedge Carex riparia, fools water cress and common reed *Phragmites australis*.

Photograph 5: Open stretch of the South Marston Brook

3.20 More shaded areas are characterised by a typical secondary woodland flora including abundant ivy *Hedera helix*, garlic mustard *Alliaria petiolata*, wood dock *Rumex sanguinea*, lesser burdock *Arctium minor* and occasional arum *Arum maculata* and wood false brome *Brachypodium sylvatica*.

Water Bodies – Standing Water

3.21 A number of ponds were identified within or just outside of the survey area: both to the north of the A420 and to the south. The habitats present vary from deep ponds with substantial bankside vegetation to ephemeral shallow ponds or recent man-made ponds. A summary of pond characteristics is provided in Table 9.6 below:

Table 1: Pond Characteristics.

Pond reference	Notes
P1	Deep pond with shallow gradient, 10 x 16 m. Heavily shaded by willow and scrub. Aquatic vegetation included *Callitriche stagnalis, Vernica beccabunga* and *Lemna minor*. Fallen branches and litter were abundant
P2	Large pond of 20 x 15 m with lush dense emergent vegetation included *Rorippa nasturtiun-aquaticum*, water crowfoot, *Glyceria maxima* and *Iris psuedacorus* with *L. minor*.
P3	Recently (part?) drained ponds with bare exposed draw-down zone evident. Some *Juncus inflexus* at former water level and part submerged tree stumps.
P4	Heavily shaded pond with many over-mature willows. Vegetation includes *Apium nodiflorum, Ranunculus sp., R. flammula*, and *Lemna* sp. Adjacent wet ditch supporting marshy grassland with abundant *Potentilla anserina*, *D. cespitosa, Carex hirta* and *Agrostis stolonifera*. Dry in summer.

Pond reference	Notes
P5	Large ornamental pond 17 x 5 m not visited in 2006. Over shaded by mature trees with little aquatic flora.
P6	Over shaded pond, 15 x 6 m. Over shaded on three sides by grown out hedgerows and ash trees. Aquatic vegetation limited to *Iris pseudacorus, Lemna and Carex otrubae*.
P7	Large pond just to the east of the River Cole. Significant marginal and emergent vegetation develops during the growing season.

3.22 Despite many of the ponds being over-shaded and/or being only of seasonal nature, due to succession/silt development, ponds at all successional stages can be important to wildlife including notable and/or protected species such as the great crested newts and other amphibians. As a result the network of ponds within the Site is considered to be of Local/Low significance.

Mature Trees

3.23 Standard trees were mainly associated with hedgerows and the South Marston Brook, ash were generally the dominant species along hedgerows, although pollarded crack willows were also frequent particularly along the course of the Brook. Overall the tree stock was in good condition and well distributed throughout the Site; enhancing the structural diversity of the Site and providing a habitat for a range of species excluded from the more intensively managed agricultural habitats. Other trees of less frequent occurrence included occasional field maple, small-leaved lime and Horse chesnut.

3.24 Over-mature and near veteran specimens were present in the form of a number of crack willows along the brook where signs of historic or even recent management were noted. As is typical and fairly common for the species, many had large hollowed out main trunks showing signs of different forms of rot, the fruiting bodies of fungus, the presence of holes used by birds and small mammals and evidence of use by saproxylic invertebrates.

3.25 More notable trees included a single veteran ash pollard (Photograph 6, TN6. Also refer to T160, Arboricultural Assessment (FPCR 2013)) of large girth and features indicative of veteran status. Veteran trees within the Site are considered to be of District/Medium significance.

Photograph 6: Ash pollard with significant hollowing of the bole.

Hedgerows

3.26 Hedgerows within the Site were of variable quality, largely due to the variation in management practice currently in operation. During surveys undertaken in 2002 and 1999, a total of hedgerows qualified as important under the Wildlife and Landscape criteria of the Hedgerow Regulations 1997 and, as a result of the review undertaken in 2006, which did not identify any significant changes, their status is likely to be maintained. Hedgerows identified as important and their HEGS grade are identified within Figure 2.

3.27 Typically hedgerows were generally dominated by hawthorn with abundant blackthorn Prunus spinosa, elm *Ulmus* spp and dog rose *Rosa canina*. The structure of hedgerows was variable, although hedgerows within the site can generally be characterised by a lack of intensive management, which has allowed them to expand in both height and width.

Arable

3.28 Arable field supported a range of crops, including wheat, linseed and turf (mainly Lolium perenne and Poa trivialis), although the latter overwhelmingly dominates arable compartments immediately adjacent to Rowborough Farm and to the south of South Marston. As a result of the intensive management of these crops, which appears to include frequent herbicide treatment, few associated species were observed. One exception was noted close to the South Marston Brook where winter flooding had prevented the germination/seeding in some areas (TN 7 where it has been colonised by a small number of persistent opportunist species including redshank *Persicaria maculosa*, greater plantain *Plantago major* and, locally, marsh foxtail.

4.0 DISCUSSION

4.1 Habitats dominating within the site were of limited interest being generally characteristic of intensively managed farmland throughout the UK. Nevertheless, some features of more interest and habitats supporting species indicative of lower levels of agricultural management were noted.

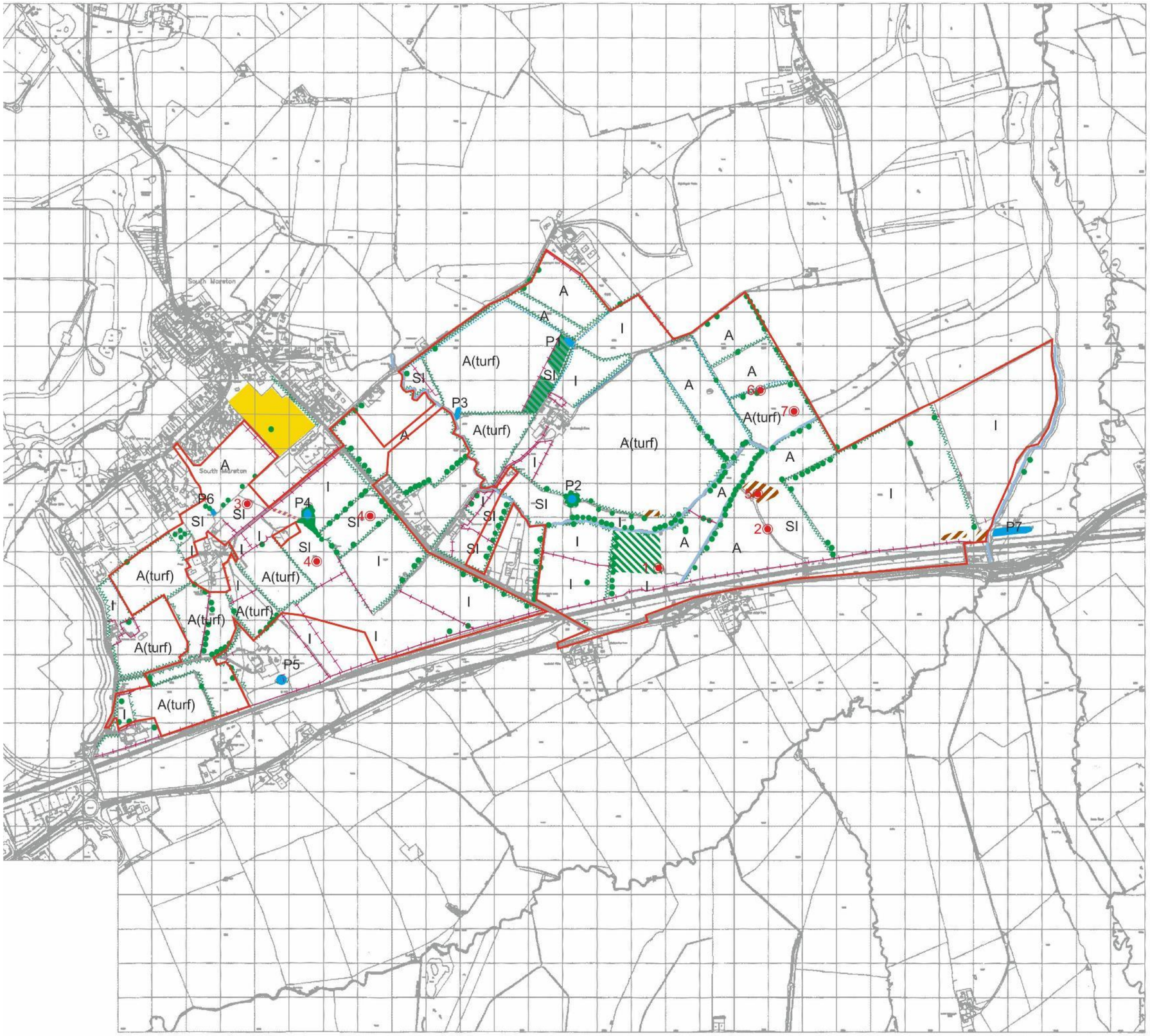
4.2 Feature of greatest interest included the South Marston Brook, which flows from the northern site boundary to its confluence with the Rover Cole in the east. This feature, in addition to its associated wet land feature, provides a relatively undisturbed and continuous habitat through otherwise heavily disturbed areas and, as a result is likely to provide an important dispersal corridor for wildlife and to forage nest and shelter commensurate with its non-statutory designation. This feature would benefit from buffering from the effects of agriculture and incorporation into the green infrastructure of any proposed development.

4.3 Hedgerows within the site are also characterised by a lack of intensive management and given their often tall, wide structure may also provide routes of movement and habitat for a range of species. They frequently support mature standard trees, which occasionally appear to be of considerable age and starting to support features characteristic of veterans. These can provide an important habitat for a range of specialist species. Where possible these should be retained or their loss compensated for to ensure that their function is maintained.

4.4 The grassland habitats within the site are of variable interest and, although largely species poor, occasionally support indicators of less intensification. Where possible these areas should form the basis for wildflower meadow creation (e.g. TN 2, 3 and 4) as the level of fertilizer input may be reduced and successful establishment of a wider range of herbaceous species more easily achieved.

4.5 The woodland resource within the site is dominated by willow coppice of varying ages. Although likely to provide a niche for some species, due to their densely-planted nature and future removal as a biofuel resource their value is likely to be transitory only and limited as a result. Where retained these features would benefit from focused management aimed at increasing their structural and species diversity both in terms of their canopy and shrub/field layers. Nevertheless, other parts of the site, such as along the South Marston Brook, do support a more typical woodland assemblage.

4.6 Other localised habitats, such as the field ponds, are also a declining biodiversity resource. Those within the site are typically, in the absence of management, in the later stages of succession, with their gradual drying through the input of organic matter and invasion by trees and scrub. This habitat type may benefit from management, incorporation into the green infrastructure and further augmentation through addition pond creation.



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- Survey Boundary
- Broad-leaved plantation woodland
- Conifer plantation woodland
- SI / I — Semi-improved/Improved grassland
- Marshy grassland
- Tall ruderal herb
- Standard tree
- Amenity grassland
- Hedgerow
- Pond (with reference)
- Watercourse
- A — Arable
- Fence
- 1 — Target note

Hallam Land Management, Taylor Wimpey & Hannick Homes

Swindon

PHASE 1 HABITAT PLAN

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Figure 1 **1796-E-A**

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Scientific	Common	Abundance
Broad-leaved woodland plantation (inc TN 1)		
Arrhenatherum elatius	False-oat grass	A
Carex acutiformis	lesser pond sedge	LD
Dryopteris filix-mas	Male fern	R
Elytrigia repens	Common couch	LA
Filipendula ulmaria	meadow sweet	O
Phalaris arundinacea	reed canary grass	LA
Poa trivialis	Rough meadow grass	A
Salix caprea	Goat willow	O
Salix fragilis	Crack willow	O
Salix viminalis	osier	D
Symphytum officinale	common comfrey	LA
Ulmus glabra	wych elm	O
Urtica dioica	nettle	A
Coniferous plantation		
Arrhenatherum elatius	False-oat grass	R
Picea abies	Norway spruce	D
Poa trivialis	Rough meadow grass	O
Ranunculus repens	Creeping buttercup	O
Urtica dioica	nettle	O
Semi-improved grassland (adj. to rail line)		
Achillea millefolium	Yarrow	O
Alopecurus pratensis	Meadow foxtail	O
Angelica sylvestris	angelica	O
Anthriscus sylvestris	Cow parsley	O
Arrhenatherum elatius	False-oat grass	D
Bellis perennis	Daisy	R
Carex flacca	Glaucus sedge	R
Carex hirta	Hairy Sedge	O
Carex otrubae	False fox sedge	R
Cerastium fontanum	Common mouse-ear	O
Conium maculata	Hemlock	LA
Dactylus glomerata	Cock's-foot	O
Deschampsia cespitosa	Tufted hair grass	O
Elytrigia repens	Common couch	LD
Equisetum arvense	Field horsetail	O
Festuca rubra	Red fescue	F
Filipendula ulmaria	Meadow sweet	R
Gallium aparine	Cleavers	R
Glechoma hederacea	Ground ivy	R
Hypericum tetragonum	Square stalked st. John's wort	R
Juncus inflexus	Hard rush	R
Lathyrus pratensis	Meadow vetchling	R
Leucanthemum vulgare	Oxeye daisy	O
Melilotus officinalis	Ribbed melilot	
Potentilla reptans	Creeping cinquefoil	O
Ranunculus acris	Meadow buttercup	O
Ranunculus repens	Creeping buttercup	F
Rubus fruticosus agg.	A bramble	O

Scientific	Common	Abundance
Symphytum officinale	common comfrey	LF
Urtica dioica	nettle	LF
Vicia sativa	Common vetch	O
Semi-improved grassland (adj track TN 2)		
Arrhenatherum elatius	False-oat grass	F
Bromus hordeaceus	Soft brome	F
Dactylus glomerata	Cock's-foot	F
Deschampsia cespitosa	Tufted hair grass	F
Festuca rubra	Red fescue	F
Galium verum	Ladies bedstraw	R
Gallium album	Hedge bedstraw	R
Glechoma hederacea	Ground ivy	O
Holcus lanatus	Yorkshire fog	F
Lotus corniculatus	Common bird's-foot trefoil	O
Medicago lupulina	Black medick	O
Poa compressa	Flattened meadow grass	R
Poa trivialis	Rough meadow grass	O
Potentilla reptans	Creeping cinquefoil	O
Ranunculus repens	Creeping buttercup	O
Senecio erucifolius	Hoary ragwort	R
Trifolium hybridum	Alsike clover	R
Trifolium repens	White clover	LA
Vicia cracca	Tufted vetch	O
Vicia sativa	Common vetch	R
Semi-improved grassland – general pasture (inc TN 3)		
Agrostis capillaris	Common bent	F
Agrostis stolonifera	Creeping bent	F
Alopecurus pratensis	Meadow foxtail	F
Alopecurus geniculatus	Marsh foxtail	R
Anthriscus sylvestris	Cow parsley	O
Arrhenatherum elatius	False-oat grass	F
Bromus hordeaceus	Soft brome	O
Capsella bursa-pastoris	Shepherd's purse	R
Carex hirta	Hairy sedge	R
Carex spicata	Spiked sedge	R
Cerastium fontanum	Common mouse-ear	O
Cerastium glomeratum	Sticky mouse-ear	R
Dactylus glomerata	Cock's-foot	F
Epilobium parviflorum	Hoary willowherb	R
Festuca arundinacea	Tall fescue	R
Festuca rubra	Red fescue	F
Geranium dissectum	Cut-leaved crane's-bill	O
Heracleum sphondylium	Hogweed	O
Holcus lanatus	Yorkshire fog	F
Hordeum secalinum	Meadow barley	R/VLF
Juncus effusus	Soft rush	R
Leucanthemum vulgare	Oxeye daisy	R
Lolium perenne	Perennial ryegrass	F
Medicago lupulina	Black medick	R

Scientific	Common	Abundance
Plantago lanceolata	Ribwort plantain	F
Plantago major	Greater plantain	O
Poa annua	Annual meadow grass	O
Poa pratensis	Smooth meadow grass	O
Poa trivialis	Rough meadow grass	F
Potentilla anserina	Silverweed	VLF
Prunella vulgaris	Selfheal	R
Ranunculus acris	Meadow buttercup	F
Ranunculus repens	Creeping buttercup	F
Rumex acetosa	Common sorrel	R
Rumex conglomeratus	Clustered dock	R
Rumex crispus	Curled dock	O
Rumex obtusifolius	Broad-leaved dock	O
Senecio erucifolius	Hoary ragwort	R
Trifolium repens	White clover	A
Trifolium pratense	Red clover	O
Urtica dioica	nettle	O
Veronica chamaedrys	Germander speedwell	R
Veronica serpyllifolia	Thyme-leaved speedwell	R
Vicia saliva	Common vetch	O
Semi-improved grassland – damp (inc TN 4)		
Agrostis stolonifera	Creeping bent	D
Alopecurus pratensis	Meadow foxtail	A
Alopecurus geniculatus	Marsh foxtail	VLA
Carex hirta	Hairy sedge	R
Cerastium fontanum	Common mouse-ear	O
Cirsium arvense	Creeping thistle	O
Glyceria fluitans	Flote grass	VLA
Holcus lanatus	Yorkshire fog	A
Juncus articulatus	Jointed rush	VLF
Juncus inflexus	Hard rush	VLF
Lathyrus pratensis	Meadow vetchling	R/VLA
Lolium perenne	Perennial ryegrass	F
Poa trivialis	Rough meadow grass	O
Potentilla anserina	Silverweed	LA
Ranunculus acris	Meadow buttercup	F
Ranunculus repens	Creeping buttercup	F
Rumex obtusifolius	Broad-leaved dock	O
Senecio erucifolius	Hoary ragwort	R
Tall Ruderal		
Anthriscus sylvestris	Cow parsley	F
Arrhenatherum elatius	False-oat grass	A
Deschampsia cespitosa	Tufted hair grass	R
Dipsacus fullonum	Teasel	O
Elytrigia repens	Common couch	F
Equisetum arvense	Field horsetail	O
Galium aparine	cleavers	A
Heracleum sphondylium	Hogweed	F
Holcus lanatus	Yorkshire fog	F
Ranunculus repens	Creeping buttercup	F

Scientific	Common	Abundance
Rubus fruticosus agg.	bramble	F
Rumex conglomeratus	Clustered dock	R
Symphytum officinale	common comfrey	VLF
Urtica dioica	nettle	A
South Marston Brook (inc margins and banks)		
Acer campestre	Field maple	O
Alliaria petiolata	Garlic mustard	F
Apium nodiflorum	fools watercress	O
Arctium minor	Lesser burdock	O
Arum maculatum	Lords and ladies	O
Brachypodium sylvatica	Wood false brome	R
Bromus sterilis	Sterile brome	O
Carex riparia	greater pond sedge	LA
Conium maculatum	Hemlock	O
Dipsacus fullonum	Teasel	O
Eleocharis palustris	Common spike rush	R
Epilobium hirsuta	Greater willowherb	F
Equisetum fluviatile	Marsh horsetail	R
Filipendula ulmaria	Meadow sweet	O
Galium aparine	Cleavers	F
Geum urbanum	Wood avens	O
Glechoma hederacea	Ground ivy	O
Glyceria fluitans	Flote grass	O
Hedera helix	Ivy	O
Humulus lupulus	Hop	R
Iris pseudacorus	yellow flag iris	LA
Lamium alba	White dead nettle	O
Lycopus europaea	Gypsywort	O
Lysimachia nummularis	Creeping jenny	R
Mentha aquatica	water mint	F
Phalaris arundinacea	Reed canary grass	O
Phragmites australis	Common reed	O
Poa trivialis	Rough meadow grass	A
Polygonum amphibium	amphibious bistort	O
Quercus robur	Pedunculate oak	O
Rorippa nasturtium -aquatica	Water cress	O
Rumex sanguineum	Wood dock	R
Salix caprea	Goat willow	O
Scrophularia auriculata	Water figwort	O
Silene dioica	Red campion	O
Tilia chordata	Small-leaved lime	R
Sparganium erectum	branched bur-reed	F
Stachys palustris,	marsh woundwort	O
Symphytum officinale	common comfrey	LF
Typha latifolia	Bulrush	LA
Urtica dioica	Nettle	A
Veronica beccabunga	brooklime	O
Pond P1		
Callitriche stagnalis	Pond water starwort	
Veronica beccabunga	brooklime	

Scientific	Common	Abundance
Lemna minor	Common duckweed	
Pond P2		
Callitriche stagnalis	Pond water starwort	
Veronica beccabunga	brooklime	
Lemna minor	Common duckweed	
Rorippa nasturtium-aquaticum	Watercress	
Ranunculus sp	water crowfoot	
Glyceria maxima	Reed sweetgrass	
Iris pseudacorus	Yellow flag iris	
Carex otrubae	False fox sedge	
Carex hirta	Hairy sedge	
Potentilla anserina	Silverweed	
Juncus articulatus	Jointed rush	
Salix fragilis	Crack willow	
Pond P3		
Juncus inflexus		
Pond P4		
Apium nodiflorum	Fools watercress	
Ranunculus sp	A water crowfoot	
R. flammula	Lesser spearwort	
Lemna sp	A duckweed	
Potentilla anserina,	silverweed	
D. cespitosa	Tufted hair grass	
Carex hirta	Hairy sedge	
Agrostis stolonifera.	Creeping bent	
Pond P6		
Iris pseudacorus	Yellow flag iris	
Carex otrubae	False fox sedge	
Arable fields (inc turf)		
Alopecurus geniculatus	Marsh foxtail	
Lolium perenne	Perennial ryegrass	
Poa trivialis	Rough meadow grass	
Poa annua	Annual meadow grass	
Agrostis stolonifera	Creeping bent	
Medicago lupulina	Black medic	
Epilobium parviflorum	Hoary willowherb	
Plantago major	Greater plantain	
Arrhenatherum elatius	False oat grass	
Linum usitatissimum	Flax/linseed	
Triticum sp	wheat	
Polygonum aviculare	knotgrass	
Polygonum maculosa	redshank	
Alopecurus myosuroides	blackgrass	